

Supplementary File

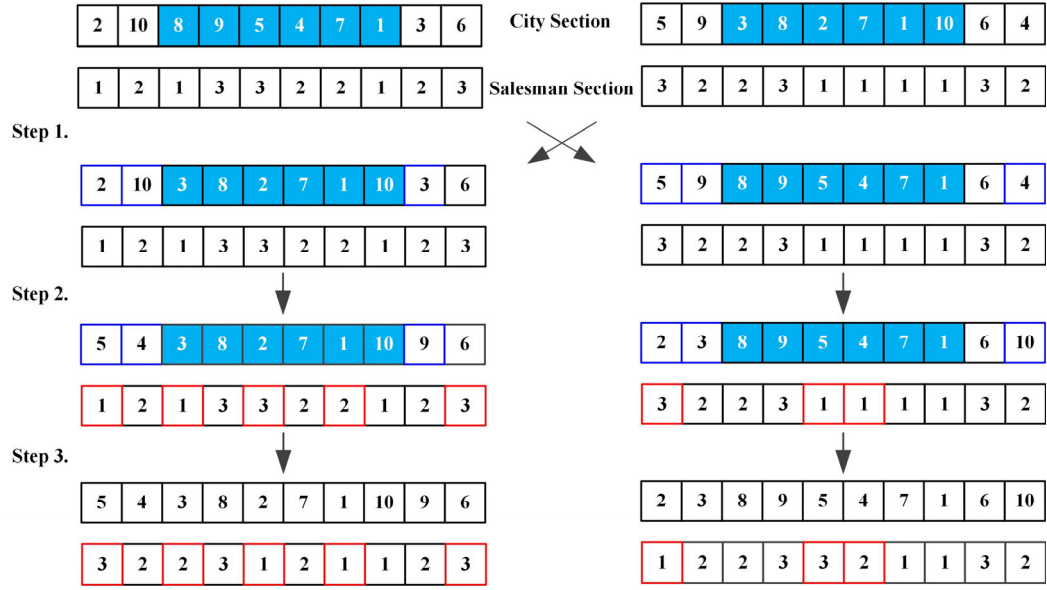


Fig. S1. Crossover process.

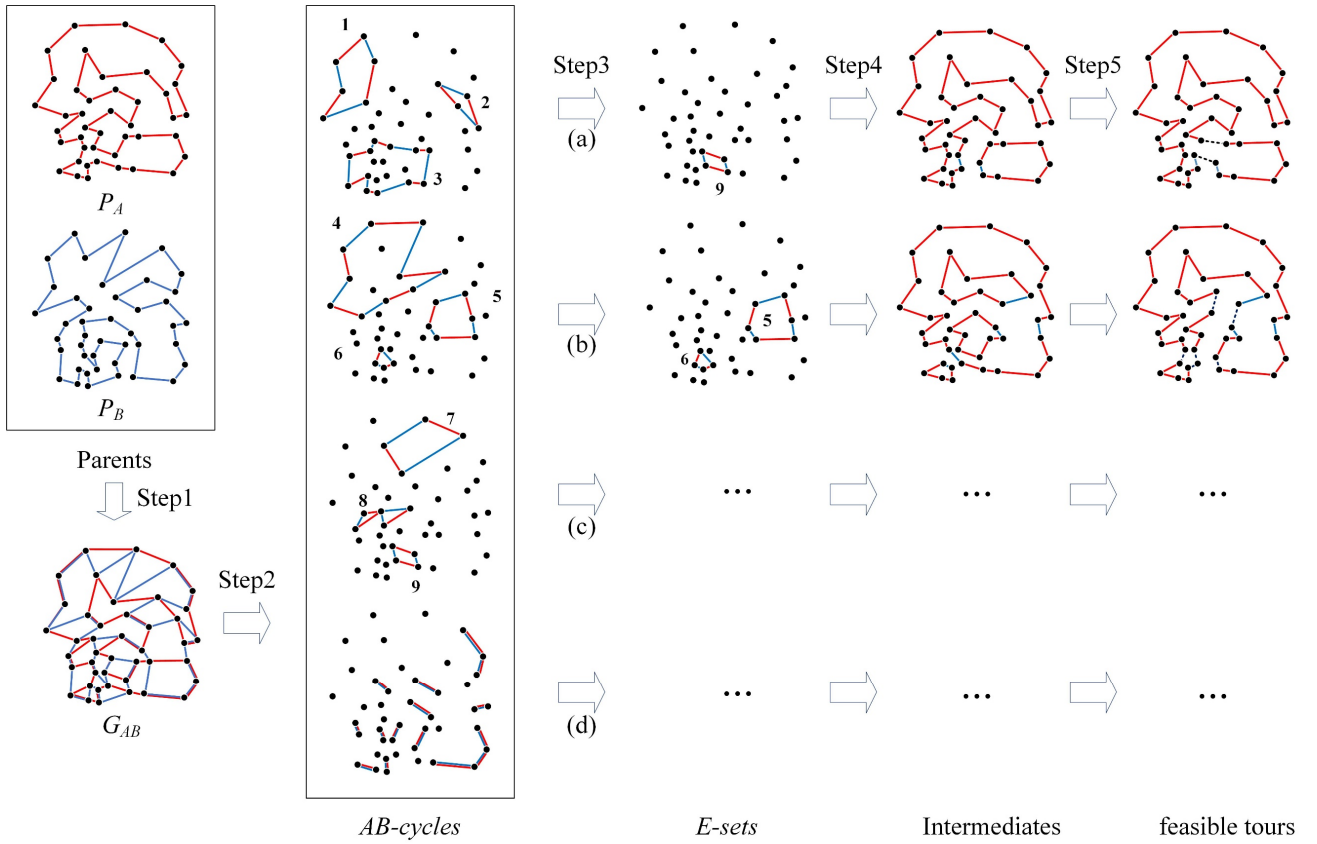


Fig. S2. EAX process

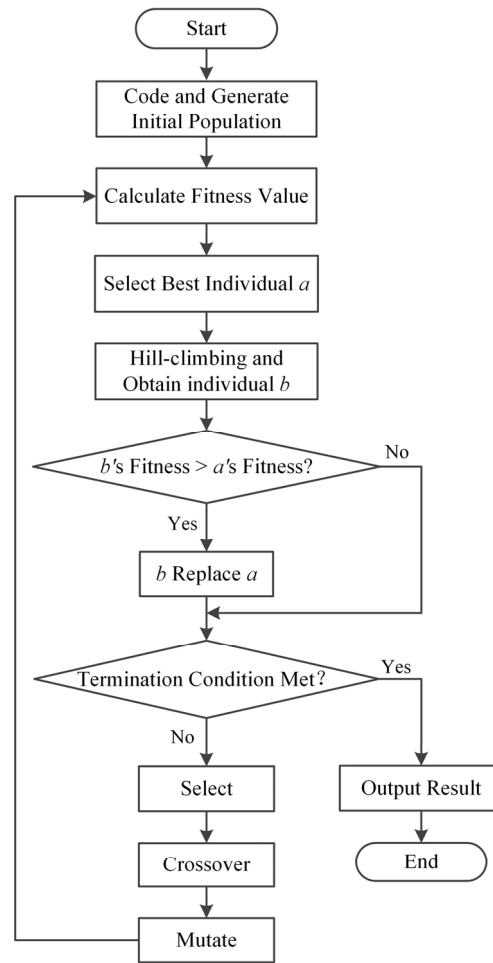


Fig. S3. Flowchart of Hill-climbing GA.

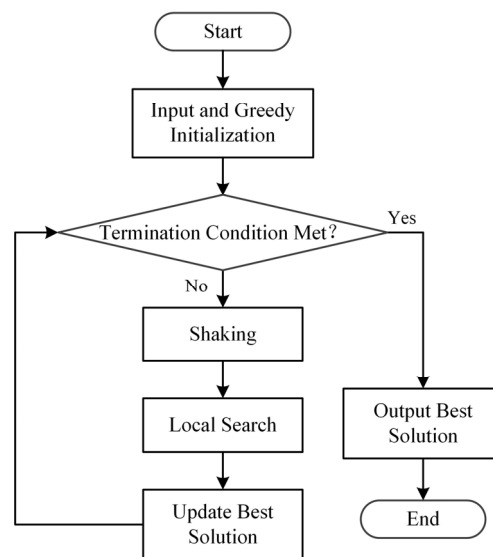


Fig. S4. Flowchart of VNS.

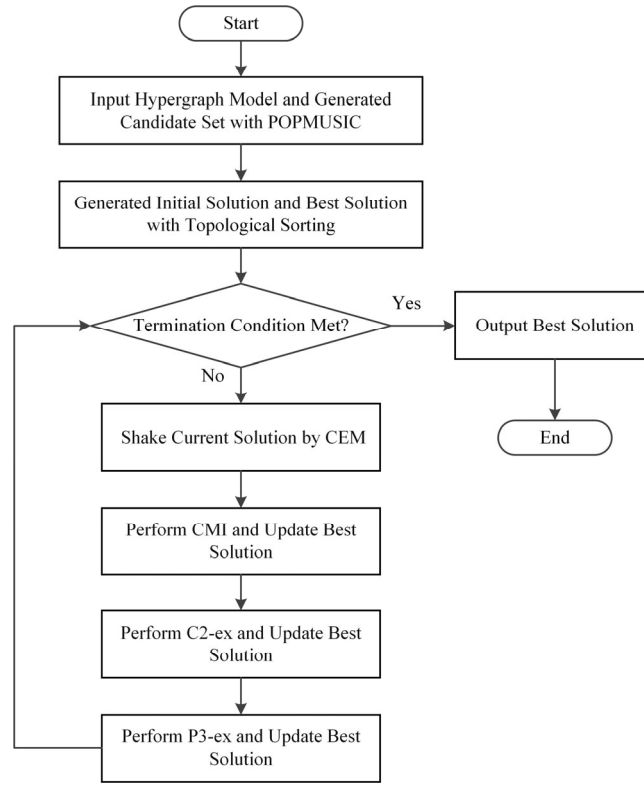


Fig. S5. Flowchart of POPMUSIC-based VNS.

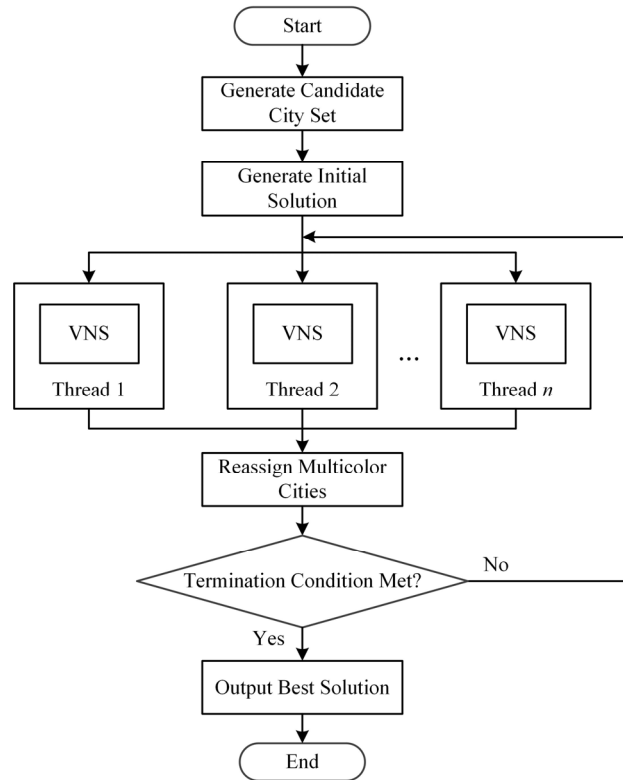


Fig. S6. Flowchart of parallel VNS.



(a)



(b)



(c)



(d)

Fig. S7. Typical multi-truss robotic systems. (a) A double-bridge waterjet cutting system [120]. (b) An automated container terminal [126]. (c) A multi-bridge crane for steel coil handling [127]. (d) The Autodesk cooperative printing system, Project Escher [119].



(a)



(b)

Fig. S8. Typical material-handling robots. (a) Point-to-point lifting AGVs [128]. (b) Multi-point case-handling AGVs [129].